PrintOnline.et v3.0 — Master Refactor Plan

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> **Domain:** printonline.et

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1. Executive Summary

1.1 What Changed from v2.0?

v2.0 (*Genesis*) established the production-grade platform with Supabase backend, better-auth, CMS, and order flow. v3.0 (*Precision*) is a **data integrity + UX polish** release focused on:

- **Pricing accuracy** — Matrix-based pricing to match client's exact catalog specifications
- **CMS professionalization** — Full UI overhaul from "AI slop" to premium big-tech aesthetic
- **Email integration** — Comprehensive notification system via Ethio Telecom SMTP
- **Order pipeline** — Strict 8-step status flow with admin controls and email notifications
- **Customer management** — Full CRUD with order history integration

1.2 Current Codebase STATE (v2.x baseline)

Layer	Status	Key Files

Layer	Status	Key Files
Database	☑ Operational	`supabase/migrations/001-009`, `supabase/seed/001-006`
Auth	☑ better-auth working	`lib/auth.ts`, `lib/auth-client.ts`
Storefront	☑ Functional	`app/(storefront)/`, `components/product/`, `components/home/`
Cart	☑ localStorage-based	`context/CartContext.tsx`
CMS	⚠ Functional but UI needs overhaul	`app/(cms)/`, `components/cms/`
Pricing	✕ Incorrect additive model	`components/product/ProductOrderForm.tsx` lines 48-90
Email	⚠ Infrastructure exists, not fully wired	`lib/email.ts`, `lib/email-template.ts`
Order Status	⚠ 8-step enum exists in DB, frontend partially synced	`types/database.ts` line 720+

1.3 Tech Stack (unchanged from v2)

Layer	Technology
Framework	Next.js 16 (App Router)
Database	Supabase PostgreSQL
Auth	better-auth
Styling	Tailwind CSS v4 + shadcn/ui
State	TanStack Query v5
Email	Nodemailer (Ethio Telecom SMTP)
Deploy	Vercel

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2. Pricing Logic Matrix Sync (P0 — Critical)

> [!CAUTION] Price is extremely sensitive in this app's context. Incorrect pricing can lead to financial loss and customer mistrust. This is the **highest priority** task in v3.0.

2.1 Problem Statement

The current pricing logic in [ProductOrderForm.tsx](file:///c:/Users/kidus/Documents/Projects/printonline-et/components/product/ProductOrderForm.tsx#L48-L90) uses an **additive model**:

```
// CURRENT (BROKEN) LOGIC — ProductOrderForm.tsx lines 48-90
const calculateUnitPrice = () => {
  let base = product.base_price || 0; // e.g. 3.50
  let additives = 0;
```

```
  // Doubles lamination/paper cost for 2-side (INCORRECT approach)
  if (
    isBothSides &&
    (opt.option_key === "lamination" || opt.option_key === "paper_thickness")
  ) {
    amount *= 2;
  }

  if (val.price_type === "override") {
    base = amount; // Override replaces base price
  } else {
    additives += amount; // Additive stacks on top
  }

  return base + additives; // Final = base + sum(additives) — WRONG
};
```

The problem: The client's catalog uses **matrix pricing** where the final price is determined by a **unique combination** of options, NOT by adding individual option prices together. For example:

- Business Cards: `1-side + 250gsm + no lamination = 3.50 ETB` (NOT `3.50 + 0 + 0`)
- Business Cards: `2-side + 300gsm + lamination = 10.00 ETB` (NOT `3.50 + 3.50 + 0.50 + 1.00`)

2.2 Solution: Pricing Matrix Lookup Table

Introduce a `product_pricing_matrix` table in the database that stores the exact price for each combination of options.

New Database Migration

```
-- supabase/migrations/010_pricing_matrix.sql
CREATE TABLE product_pricing_matrix (
  id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
  product_id UUID NOT NULL REFERENCES products(id) ON DELETE CASCADE,
  matrix_key TEXT NOT NULL,          -- e.g. "front_only|250gsm|none"
  matrix_label TEXT,                 -- e.g. "1 Side Print, 250gsm, No Lamination"
  price DECIMAL(12,2) NOT NULL,      -- The exact final unit price in ETB
  is_active BOOLEAN DEFAULT true,
  created_at TIMESTAMPTZ DEFAULT NOW(),
  UNIQUE(product_id, matrix_key)
);

CREATE INDEX idx_pricing_matrix_product ON product_pricing_matrix(product_id);
CREATE INDEX idx_pricing_matrix_key ON product_pricing_matrix(product_id,
matrix_key);
```

New Frontend Pricing Logic

```
// components/product/ProductOrderForm.tsx — REPLACEMENT calculateUnitPrice
const calculateUnitPrice = () => {
  // 1. Determine which option keys form the matrix for this product
  const matrixOptionKeys = [
    "print_sides",
    "paper_thickness",
    "lamination",
    "size",
    "pocket",
  ];
  const matrixParts: string[] = [];

  // Build the key in the order the matrix expects
  for (const optKey of matrixOptionKeys) {
    const opt = product.product_options?.find((o) => o.option_key === optKey);
    if (!opt) continue;
    const selectedValId = selections[opt.id];
    if (!selectedValId) continue;
    const val = opt.product_option_values?.find((v) => v.id === selectedValId);
    if (val) matrixParts.push(val.value);
  }

  const matrixKey = matrixParts.join("|");

  // 2. Lookup in pricing matrix (fetched alongside product data)
  const matrixEntry = product.pricing_matrix?.find(
    (m) => m.matrix_key === matrixKey,
  );

  if (matrixEntry) {
    return matrixEntry.price; // Exact matrix price — no additives!
  }

  // 3. Fallback to base_price if no matrix match
  return product.base_price || 0;
};
```

2.3 Product-Specific Pricing Matrices

2.3.1 Business Cards

Matrix factors: `print_sides × paper_thickness × lamination`

Print Sides	Paper	Lamination	Price (ETB/card)
`front_only`	`250gsm`	`none`	**3.50**
`front_only`	`250gsm`	`matte` or `glossy`	**4.50**
`front_only`	`300gsm`	`none`	**4.00**

Print Sides	Paper	Lamination	Price (ETB/card)
`front_only`	`300gsm`	`matte` or `glossy`	**5.00**
`both_sides`	`250gsm`	`none`	**7.00**
`both_sides`	`250gsm`	`matte` or `glossy`	**9.00**
`both_sides`	`300gsm`	`none`	**8.00**
`both_sides`	`300gsm`	`matte` or `glossy`	**10.00**

matrix_key format: `{print_sides}|{paper_thickness}|{lamination}`

```sql
-- supabase/seed/007_pricing_matrix.sql — Business Cards
INSERT INTO product_pricing_matrix (product_id, matrix_key, matrix_label, price)
VALUES
  ((SELECT id FROM products WHERE slug='business-cards'),
'front_only|250gsm|none', '1 Side, 250gsm, No Lamination', 3.50),
  ((SELECT id FROM products WHERE slug='business-cards'),
'front_only|250gsm|matte', '1 Side, 250gsm, Matte Lamination', 4.50),
  ((SELECT id FROM products WHERE slug='business-cards'),
'front_only|250gsm|glossy', '1 Side, 250gsm, Glossy Lamination', 4.50),
  ((SELECT id FROM products WHERE slug='business-cards'),
'front_only|300gsm|none', '1 Side, 300gsm, No Lamination', 4.00),
  ((SELECT id FROM products WHERE slug='business-cards'),
'front_only|300gsm|matte', '1 Side, 300gsm, Matte Lamination', 5.00),
  ((SELECT id FROM products WHERE slug='business-cards'),
'front_only|300gsm|glossy', '1 Side, 300gsm, Glossy Lamination', 5.00),
  ((SELECT id FROM products WHERE slug='business-cards'),
'both_sides|250gsm|none', '2 Sides, 250gsm, No Lamination', 7.00),
  ((SELECT id FROM products WHERE slug='business-cards'),
'both_sides|250gsm|matte', '2 Sides, 250gsm, Matte Lamination', 9.00),
  ((SELECT id FROM products WHERE slug='business-cards'),
'both_sides|250gsm|glossy', '2 Sides, 250gsm, Glossy Lamination', 9.00),
  ((SELECT id FROM products WHERE slug='business-cards'),
'both_sides|300gsm|none', '2 Sides, 300gsm, No Lamination', 8.00),
  ((SELECT id FROM products WHERE slug='business-cards'),
'both_sides|300gsm|matte', '2 Sides, 300gsm, Matte Lamination', 10.00),
  ((SELECT id FROM products WHERE slug='business-cards'),
'both_sides|300gsm|glossy', '2 Sides, 300gsm, Glossy Lamination', 10.00);
```

2.3.2 Flyers

Matrix factors: `print_sides × paper_thickness × lamination`

> [!IMPORTANT] The user explicitly provided this flyer pricing matrix. However, the DB seed data ([004_product_options_part2.sql](file:///c:/Users/kidus/Documents/Projects/printonline-et/supabase/seed/004_product_options_part2.sql#L224-L228)) shows a DIFFERENT, more complex pricing based on size × paper × sides. The user's provided data takes precedence.

> **CRITICAL:** The flyer paper_thickness options in DB are currently `80gsm` and `150gsm` (lines 203-222 of seed file). These MUST be updated to `250gsm` and `300gsm` to match the user's specs. Additionally, a `lamination` option must be ADDED to flyers — it doesn't exist in the current seed data.

Print Sides	Paper	Lamination	Price (ETB)
front_only	250gsm	none	**3.50**
front_only	250gsm	matte/glossy	**4.50**
front_only	300gsm	none	**4.00**
front_only	300gsm	matte/glossy	**5.00**
both_sides	250gsm	none	**7.00**
both_sides	250gsm	matte/glossy	**9.00**
both_sides	300gsm	none	**8.00**
both_sides	300gsm	matte/glossy	**10.00**

Required DB changes for Flyers:

1. UPDATE `product_option_values` for flyer `paper_thickness`: change `80gsm` → `250gsm`, `150gsm` → `300gsm`
2. INSERT new `product_options` row: `lamination` option for flyers
3. INSERT `product_option_values` for flyer lamination: `none`, `matte`, `glossy`

```
-- Flyers: Update paper thickness values
UPDATE product_option_values SET value='250gsm', label='250gsm'
WHERE option_id = (SELECT id FROM product_options WHERE product_id = (SELECT id
FROM products WHERE slug='flyers') AND option_key='paper_thickness') AND
value='80gsm';

UPDATE product_option_values SET value='300gsm', label='300gsm'
WHERE option_id = (SELECT id FROM product_options WHERE product_id = (SELECT id
FROM products WHERE slug='flyers') AND option_key='paper_thickness') AND
value='150gsm';

-- Flyers: Add lamination option
INSERT INTO product_options (product_id, option_key, option_label, field_type,
is_required, display_order)
VALUES ((SELECT id FROM products WHERE slug='flyers'), 'lamination', 'Lamination',
'radio', true, 4);

INSERT INTO product_option_values (option_id, value, label, display_order,
is_default)
VALUES
  ((SELECT id FROM product_options WHERE product_id=(SELECT id FROM products WHERE
slug='flyers') AND option_key='lamination'), 'none', 'None', 1, true),
  ((SELECT id FROM product_options WHERE product_id=(SELECT id FROM products WHERE
slug='flyers') AND option_key='lamination'), 'matte', 'Matte', 2, false),
  ((SELECT id FROM product_options WHERE product_id=(SELECT id FROM products WHERE
```

```
slug='flyers') AND option_key='lamination'), 'glossy', 'Glossy', 3, false);

-- Flyers pricing matrix
INSERT INTO product_pricing_matrix (product_id, matrix_key, matrix_label, price)
VALUES
  ((SELECT id FROM products WHERE slug='flyers'), 'front_only|250gsm|none', '1
Side, 250gsm, No Lamination', 3.50),
  ((SELECT id FROM products WHERE slug='flyers'), 'front_only|250gsm|matte', '1
Side, 250gsm, Matte', 4.50),
  ((SELECT id FROM products WHERE slug='flyers'), 'front_only|250gsm|glossy', '1
Side, 250gsm, Glossy', 4.50),
  ((SELECT id FROM products WHERE slug='flyers'), 'front_only|300gsm|none', '1
Side, 300gsm, No Lamination', 4.00),
  ((SELECT id FROM products WHERE slug='flyers'), 'front_only|300gsm|matte', '1
Side, 300gsm, Matte', 5.00),
  ((SELECT id FROM products WHERE slug='flyers'), 'front_only|300gsm|glossy', '1
Side, 300gsm, Glossy', 5.00),
  ((SELECT id FROM products WHERE slug='flyers'), 'both_sides|250gsm|none', '2
Sides, 250gsm, No Lamination', 7.00),
  ((SELECT id FROM products WHERE slug='flyers'), 'both_sides|250gsm|matte', '2
Sides, 250gsm, Matte', 9.00),
  ((SELECT id FROM products WHERE slug='flyers'), 'both_sides|250gsm|glossy', '2
Sides, 250gsm, Glossy', 9.00),
  ((SELECT id FROM products WHERE slug='flyers'), 'both_sides|300gsm|none', '2
Sides, 300gsm, No Lamination', 8.00),
  ((SELECT id FROM products WHERE slug='flyers'), 'both_sides|300gsm|matte', '2
Sides, 300gsm, Matte', 10.00),
  ((SELECT id FROM products WHERE slug='flyers'), 'both_sides|300gsm|glossy', '2
Sides, 300gsm, Glossy', 10.00);
```

2.3.3 Folders

Matrix factors: `print_sides × lamination × pocket`

> From seed comments ([003_product_options_part1.sql lines 640-642](file:///c:/Users/kidus/Documents/Projects/printonline-et/supabase/seed/003_product_options_part1.sql#L640-L642)):

Print Sides	Pocket	Price (ETB/pc)
`front_only`	`right_side` (1 pocket)	**350.00**
`front_only`	`left_right` (2 pockets)	**380.00**
`both_sides`	`right_side` (1 pocket)	**400.00**
`both_sides`	`left_right` (2 pockets)	**450.00**

> [!NOTE] Lamination type (matte/glossy) does NOT affect folder price — it's always included.

```
-- Folders pricing matrix
INSERT INTO product_pricing_matrix (product_id, matrix_key, matrix_label, price)
```

```
VALUES
  ((SELECT id FROM products WHERE slug='folders'), 'front_only|matte|right_side',
'Front, Matte, 1 Pocket', 350.00),
  ((SELECT id FROM products WHERE slug='folders'), 'front_only|glossy|right_side',
'Front, Glossy, 1 Pocket', 350.00),
  ((SELECT id FROM products WHERE slug='folders'), 'front_only|matte|left_right',
'Front, Matte, 2 Pockets', 380.00),
  ((SELECT id FROM products WHERE slug='folders'), 'front_only|glossy|left_right',
'Front, Glossy, 2 Pockets', 380.00),
  ((SELECT id FROM products WHERE slug='folders'), 'both_sides|matte|right_side',
'Both, Matte, 1 Pocket', 400.00),
  ((SELECT id FROM products WHERE slug='folders'), 'both_sides|glossy|right_side',
'Both, Glossy, 1 Pocket', 400.00),
  ((SELECT id FROM products WHERE slug='folders'), 'both_sides|matte|left_right',
'Both, Matte, 2 Pockets', 450.00),
  ((SELECT id FROM products WHERE slug='folders'), 'both_sides|glossy|left_right',
'Both, Glossy, 2 Pockets', 450.00);
```

2.3.4 Posters

Matrix factors: `paper_thickness × lamination`

> From seed comments ([004_product_options_part2.sql lines 515-517](file:///c:/Users/kidus/Documents/Projects/printonline-et/supabase/seed/004_product_options_part2.sql#L515-L517)):

Paper	Lamination	Price (ETB/pc)
150gsm	none	**54.00**
150gsm	matte/glossy	**70.00**
250gsm	none	**70.00**
250gsm	matte/glossy	**90.00**

```
INSERT INTO product_pricing_matrix (product_id, matrix_key, matrix_label, price)
VALUES
  ((SELECT id FROM products WHERE slug='posters'), '150gsm|none', '150gsm, No
Lamination', 54.00),
  ((SELECT id FROM products WHERE slug='posters'), '150gsm|matte', '150gsm,
Matte', 70.00),
  ((SELECT id FROM products WHERE slug='posters'), '150gsm|glossy', '150gsm,
Glossy', 70.00),
  ((SELECT id FROM products WHERE slug='posters'), '250gsm|none', '250gsm, No
Lamination', 70.00),
  ((SELECT id FROM products WHERE slug='posters'), '250gsm|matte', '250gsm,
Matte', 90.00),
  ((SELECT id FROM products WHERE slug='posters'), '250gsm|glossy', '250gsm,
Glossy', 90.00);
```

2.3.5 Paper Sticker Sheets

Matrix factors: `size × lamination`

> [!IMPORTANT] A `lamination` option MUST be added to paper sticker sheets — check if it exists in [005_product_options_part3.sql](file:///c:/Users/kidus/Documents/Projects/printonline-et/supabase/seed/005_product_options_part3.sql). The current seed may only have a `size` option. The `min_order_quantity` is correctly set to 24 in [002_products.sql line 301](file:///c:/Users/kidus/Documents/Projects/printonline-et/supabase/seed/002_products.sql#L301).

Size	Lamination	Price (ETB/sheet)
a4	none	**50.00**
a4	matte/glossy	**60.00**
a5	none	**25.00**
a5	matte/glossy	**30.00**
a6	none	**13.00**
a6	matte/glossy	**15.00**

```sql
-- Add lamination option to paper sticker sheets (if not already present)
INSERT INTO product_options (product_id, option_key, option_label, field_type,
is_required, display_order)
VALUES ((SELECT id FROM products WHERE slug='paper-sticker-sheets'), 'lamination',
'Lamination', 'radio', true, 2);

INSERT INTO product_option_values (option_id, value, label, display_order,
is_default) VALUES
  ((SELECT id FROM product_options WHERE product_id=(SELECT id FROM products WHERE
slug='paper-sticker-sheets') AND option_key='lamination'), 'none', 'None', 1,
true),
  ((SELECT id FROM product_options WHERE product_id=(SELECT id FROM products WHERE
slug='paper-sticker-sheets') AND option_key='lamination'), 'matte', 'Matte', 2,
false),
  ((SELECT id FROM product_options WHERE product_id=(SELECT id FROM products WHERE
slug='paper-sticker-sheets') AND option_key='lamination'), 'glossy', 'Glossy', 3,
false);

-- Paper Sticker Sheets pricing matrix
INSERT INTO product_pricing_matrix (product_id, matrix_key, matrix_label, price)
VALUES
  ((SELECT id FROM products WHERE slug='paper-sticker-sheets'), 'a4|none', 'A4, No
Lamination', 50.00),
  ((SELECT id FROM products WHERE slug='paper-sticker-sheets'), 'a4|matte', 'A4,
Matte', 60.00),
  ((SELECT id FROM products WHERE slug='paper-sticker-sheets'), 'a4|glossy', 'A4,
Glossy', 60.00),
  ((SELECT id FROM products WHERE slug='paper-sticker-sheets'), 'a5|none', 'A5, No
Lamination', 25.00),
```

```
  ((SELECT id FROM products WHERE slug='paper-sticker-sheets'), 'a5|matte', 'A5,
Matte', 30.00),
  ((SELECT id FROM products WHERE slug='paper-sticker-sheets'), 'a5|glossy', 'A5,
Glossy', 30.00),
  ((SELECT id FROM products WHERE slug='paper-sticker-sheets'), 'a6|none', 'A6, No
Lamination', 13.00),
  ((SELECT id FROM products WHERE slug='paper-sticker-sheets'), 'a6|matte', 'A6,
Matte', 15.00),
  ((SELECT id FROM products WHERE slug='paper-sticker-sheets'), 'a6|glossy', 'A6,
Glossy', 15.00);
```

2.3.6 Bookmarks

Matrix factors: `print_sides × lamination`

Print Side	Lamination	Price (ETB/pc)
`front_only`	`none`	**10.00**
`front_only`	`matte/glossy`	**12.00**
`both_sides`	`none`	**20.00**
`both_sides`	`matte/glossy`	**24.00**

```
INSERT INTO product_pricing_matrix (product_id, matrix_key, matrix_label, price)
VALUES
  ((SELECT id FROM products WHERE slug='bookmarks'), 'front_only|none', 'Front
Only, No Lamination', 10.00),
  ((SELECT id FROM products WHERE slug='bookmarks'), 'front_only|matte', 'Front
Only, Matte', 12.00),
  ((SELECT id FROM products WHERE slug='bookmarks'), 'front_only|glossy', 'Front
Only, Glossy', 12.00),
  ((SELECT id FROM products WHERE slug='bookmarks'), 'both_sides|none', 'Both
Sides, No Lamination', 20.00),
  ((SELECT id FROM products WHERE slug='bookmarks'), 'both_sides|matte', 'Both
Sides, Matte', 24.00),
  ((SELECT id FROM products WHERE slug='bookmarks'), 'both_sides|glossy', 'Both
Sides, Glossy', 24.00);
```

2.4 Price Display UX Fix

Current issue: Options display `+ETB 5.00` on radio/select labels (lines 231-235 and 263-266 of [ProductOrderForm.tsx](file:///c:/Users/kidus/Documents/Projects/printonline-et/components/product/ProductOrderForm.tsx)), which misleads users into thinking it's an additional cost on top.

Fix: Remove the `+` sign from all option price displays. For matrix-priced products, consider hiding per-option prices entirely and showing only the computed total.

```
// Line 233 (select items)
- +ETB {val.price_amount}
+ ETB {val.price_amount}

// Line 265 (radio labels)
- +ETB {val.price_amount}
+ ETB {val.price_amount}
```

2.5 Implementation Checklist

- ☐ Create `supabase/migrations/010_pricing_matrix.sql` — new table
- ☐ Create `supabase/seed/007_pricing_matrix.sql` — seed all 6 product matrices
- ☐ Update flyer options: `paper_thickness` from `80gsm/150gsm` → `250gsm/300gsm`, add `lamination`
- ☐ Add `lamination` option to paper sticker sheets
- ☐ Update [types/database.ts](file:///c:/Users/kidus/Documents/Projects/printonline-et/types/database.ts) — regenerate or manually add `product_pricing_matrix`
- ☐ Add `pricing_matrix` to `ProductWithDetails` type in [types/index.ts](file:///c:/Users/kidus/Documents/Projects/printonline-et/types/index.ts)
- ☐ Update [hooks/data/useProduct.ts](file:///c:/Users/kidus/Documents/Projects/printonline-et/hooks/data/useProduct.ts) — include `pricing_matrix` in select
- ☐ Refactor [ProductOrderForm.tsx](file:///c:/Users/kidus/Documents/Projects/printonline-et/components/product/ProductOrderForm.tsx) — replace `calculateUnitPrice()` entirely
- ☐ Remove `+` sign from option price labels (lines 233, 265)
- ☐ Test all 6 products end-to-end

2.6 Files to Modify

File	Action	Description
`supabase/migrations/010_pricing_matrix.sql`	**CREATE**	New pricing matrix table
`supabase/seed/007_pricing_matrix.sql`	**CREATE**	All product pricing data + option fixes
`types/database.ts`	**REGENERATE**	Add `product_pricing_matrix` type
`types/index.ts`	**MODIFY**	Add `pricing_matrix` to `ProductWithDetails`
`hooks/data/useProduct.ts`	**MODIFY**	Include pricing_matrix in product query
`components/product/ProductOrderForm.tsx`	**MAJOR REFACTOR**	Replace pricing logic, remove `+` signs

3. Storefront UX Patches

3.1 Terms & Conditions on Checkout

Requirement: Add a mandatory "Terms and Conditions" checkbox on the final ordering step.

Current state: The `orders` table already has `terms_accepted` and `terms_accepted_at` columns (from [006_orders.sql](file:///c:/Users/kidus/Documents/Projects/printonline-et/supabase/migrations/006_orders.sql)). Content exists at `content/`.

Implementation:

1. Create `components/checkout/TermsCheckbox.tsx` — checkbox + expandable T&C modal
2. Modify [app/order-summary/page.tsx](file:///c:/Users/kidus/Documents/Projects/printonline-et/app/order-summary/page.tsx) — add checkbox before "Place Order"
3. **Disable** "Place Order" until checkbox is checked
4. On order submission, set `terms_accepted: true` and `terms_accepted_at: new Date()`

Files:

- `components/checkout/TermsCheckbox.tsx` — **CREATE**
- `app/order-summary/page.tsx` — **MODIFY** (add T&C gate)

3.2 Out of Stock Badging

Requirement: Dynamic "Out of Stock" label for ready-made Promotional Items.

Current state: `products.stock_status` field exists. [ProductOrderForm.tsx line 45](file:///c:/Users/kidus/Documents/Projects/printonline-et/components/product/ProductOrderForm.tsx#L45) already checks and disables the cart button.

Implementation:

1. Add "Out of Stock" badge overlay on [ProductCard.tsx](file:///c:/Users/kidus/Documents/Projects/printonline-et/components/shared/ProductCard.tsx)
2. Add banner on [ProductInfo.tsx](file:///c:/Users/kidus/Documents/Projects/printonline-et/components/product/ProductInfo.tsx)
3. Badge style: semi-transparent red overlay or ribbon

Files:

- `components/shared/ProductCard.tsx` — **MODIFY**
- `components/product/ProductInfo.tsx` — **MODIFY**

3.3 Auth Emails (Welcome + Password Reset)

Requirement: Fix email dispatches for Account Creation and Password Reset.

Current state:

- [lib/email.ts](file:///c:/Users/kidus/Documents/Projects/printonline-et/lib/email.ts) — Nodemailer transport exists
- [lib/email-template.ts](file:///c:/Users/kidus/Documents/Projects/printonline-et/lib/email-template.ts) — HTML template builder exists
- SMTP configured in `.env.local`

Files:

- `lib/email-templates/welcome.ts` — **CREATE**
- `lib/email-templates/password-reset.ts` — **CREATE**
- `lib/auth.ts` — **MODIFY** (add email hooks for better-auth)
- **Leverage [/email-integration](file:///c:/Users/kidus/Documents/Projects/printonline-et/.agent/workflows/email-integration.md) workflow**

---

4. File Upload Overhaul

4.1 Multiple File Upload

Current state: Single file upload in [ProductOrderForm.tsx line 40, 101-108](file:///c:/Users/kidus/Documents/Projects/printonline-et/components/product/ProductOrderForm.tsx#L40):

```
const [designFile, setDesignFile] = useState<File | null>(null);
```

Implementation:

1. Change state to `File[]` array
2. Add `multiple` attribute to `<input type="file">`
3. Enforce: **max 4 files**, reasonable size limit per file (e.g., 10MB each)
4. Display all file names in UI
5. Update [CartItem type](file:///c:/Users/kidus/Documents/Projects/printonline-et/context/CartContext.tsx#L18-L41) for `designFileNames: string[]`
6. Upload to Supabase Storage on order submission
7. Store file references in `order_items`

4.2 Label Change

Change [line 430](file:///c:/Users/kidus/Documents/Projects/printonline-et/components/product/ProductOrderForm.tsx#L430):

```
- <span>Need Design Help?</span>
+ <span>I don't have a Design</span>
```

Files:

- `components/product/ProductOrderForm.tsx` — **MODIFY**
- `context/CartContext.tsx` — **MODIFY** (update CartItem type)
- `app/order-summary/page.tsx` — **MODIFY** (handle multiple files)

---

5. Quantity Enforcement

5.1 Minimum Quantity per Product

Current state: The `products.min_order_quantity` values are set correctly in [002_products.sql](file:///c:/Users/kidus/Documents/Projects/printonline-et/supabase/seed/002_products.sql):

Product	min_order_quantity
Business Cards	50
Letterhead	50
Envelopes	25
Folders	50
Flyers	50
Brochures	50
Posters	50
Bookmarks	50
All Booklets	50
Notebooks	50
Paper Sticker Sheets	24
Premium Gift Bags	50
Certificate Paper	1

Implementation:

1. Quantity selector in [ProductOrderForm.tsx lines 357-379](file:///c:/Users/kidus/Documents/Projects/printonline-et/components/product/ProductOrderForm.tsx#L357-L379) — **start from `product.min_order_quantity`**
2. Default quantity = `min_order_quantity`
3. Display "Minimum order: X pieces" near selector
4. **Remove quantity selector from cart page** — quantity is final from product config

5.2 Current Quantity Logic Issue

The current code (line 365-367) hardcodes quantity options by product slug:

```
{(product.slug === "business-cards"
  ? [50, 100, 250, 500, 1000, 2000, 5000]
  : [1, 5, 10, 25, 50, 100, 250, 500, 1000]
).map(...)}
```

This must be replaced with dynamic options that respect `min_order_quantity`.

Files:

- `components/product/ProductOrderForm.tsx` — **MODIFY**
- `components/cart/CartItem.tsx` — **MODIFY** (remove quantity +/-)
- `components/layout/CartDrawer.tsx` — **MODIFY** (remove quantity editing)

---

6. Rush Production Fix

6.1 Rush Price Per Order (Not Per Piece)

Current state: [ProductOrderForm.tsx lines 93-94](file:///c:/Users/kidus/Documents/Projects/printonline-et/components/product/ProductOrderForm.tsx#L93-L94):

```
const priorityPrice = productionPriority === "rush" ? 500 : 0;
const totalPrice = unitPrice * quantity + priorityPrice;
```

This adds rush as flat 500 once — **appears correct at product level**. BUT: `priorityPrice` is stored in cart item (line 180) and the [cart total computation](file:///c:/Users/kidus/Documents/Projects/printonline-et/context/CartContext.tsx#L146-L151) only sums `unitPrice * quantity`, so rush price is **lost** in the overall cart total.

Fix:

1. Update `getCartTotal()` in [CartContext.tsx](file:///c:/Users/kidus/Documents/Projects/printonline-et/context/CartContext.tsx#L146-L151) to include `priorityPrice`:

```
const getCartTotal = useCallback(() => {
  return cart.reduce(
    (total, item) =>
      total + item.unitPrice * item.quantity + (item.priorityPrice || 0),
    0,
  );
}, [cart]);
```

2. Update CartItem interface to include `priorityPrice: number`
3. Standard production = FREE (0 ETB) ☑ already correct

Files:

- `context/CartContext.tsx` — **MODIFY**
- `app/order-summary/page.tsx` — **VERIFY**

---

7. Search Bar Functionalization

Requirement: Fully functional premium search in the navbar.

Current state:

- `components/shared/SearchBar.tsx` — exists
- `app/(storefront)/search/page.tsx` — search results page exists
- Full-text search GIN index on `products` table ([002_products.sql lines 424-425](file:///c:/Users/kidus/Documents/Projects/printonline-et/supabase/migrations/002_products.sql))
- `hooks/data/useSearch.ts` — search hook exists
- `hooks/ui/useDebounce.ts` — debounce hook exists

Implementation:

1. **Debounced input** → live dropdown as user types (5-8 results with thumbnails)
2. **Full results page** → Enter/click "View All" → `/search?q=...`
3. **Premium styling** — glassmorphism dropdown, Framer Motion animations
4. **Keyboard**: `Cmd+K` to focus search
5. **Mobile** — full-width overlay

Files:

- `components/shared/SearchBar.tsx` — **MAJOR REFACTOR**
- `components/Header.tsx` — **MODIFY** (ensure integration)

---

8. CMS UI/UX Major Overhaul

> [!IMPORTANT] The entire CMS needs a shift from "AI slop" to premium, big-tech aesthetic. Focus: typography weight, animations, layout professionalism.

8.1 Sidebar Refactor

Current state: [CMSSidebar.tsx](file:///c:/Users/kidus/Documents/Projects/printonline-et/components/cms/layout/CMSSidebar.tsx) — "P" square logo + "PrintOnline / PANA ADMIN" text.

Changes:

1. Replace "P" box with `public/nav-logo.png` (`<Image src="/nav-logo.png" ... />`)
2. Remove text ("PrintOnline" / "PANA ADMIN") — logo only
3. Remove Settings nav link (lines 168-173)
4. Mark Products and Categories as "Under Maintenance" with visual badge

Current nav array (lines 75-81):

```
const navigation = [
  { href: "/cms", icon: BarChart3, label: "Overview" },
  { href: "/cms/orders", icon: ShoppingCart, label: "Orders" },
  { href: "/cms/products", icon: Package, label: "Products" }, // → Maintenance
  { href: "/cms/customers", icon: Users, label: "Customers" },
  { href: "/cms/categories", icon: Layers, label: "Categories" }, // → Maintenance
];
```

8.2 Font Weight Overhaul

Remove excessive `font-bold` and `uppercase tracking-widest` across all CMS components.

Guidelines:

- Page titles: `font-semibold`
- Subtitles/labels: `font-medium`
- Body text: `font-normal`
- Badges/tags: `font-medium`
- Button text: `font-semibold`

8.3 Animation Overhaul

Remove all aggressive spring/bounce/rotate animations across CMS.

Replace with:

- Subtle `ease-out` transitions (150-300ms max)
- Gentle fade-in page transitions
- Subtle shadow elevation on hover (no scale)
- No `rotate-3`, `scale-[1.02]`, heavy `animate-in` effects

8.4 Dashboard Redesign

Redesign [cms/page.tsx](<file:///c:/Users/kidus/Documents/Projects/printonline-et/app/(cms)/cms/page.tsx>):

- Clean minimal stats cards
- Recent orders snippet table
- Quick action buttons
- Logo in header area

8.5 CMS Files to Modify

File	Change
[CMSSidebar.tsx](file:///c:/Users/kidus/Documents/Projects/printonline-et/components/cms/layout/CMSSidebar.tsx)	Logo, remove settings, maintenance
[CMSHeader.tsx](file:///c:/Users/kidus/Documents/Projects/printonline-et/components/cms/layout/CMSHeader.tsx)	Remove text, keep logo
[cms/page.tsx](<file:///c:/Users/kidus/Documents/Projects/printonline-et/app/(cms)/cms/page.tsx>)	Premium dashboard
All `components/cms/**`	Font weight + animation audit

---

9. Order Management Refactor

9.1 Order List Polish

File: [OrderList.tsx](file:///c:/Users/kidus/Documents/Projects/printonline-et/components/cms/orders/OrderList.tsx)

- Add columns: order date, customer name, item count, total, status
- Ensure **CSV export works**
- 8-step status badge colors

9.2 Order Actions

Cancel Order:

1. Functional cancel button
2. Reusable `CMSConfirmDialog` confirmation
3. Update status to `cancelled`, append to `status_history`
4. Fire cancellation email
5. **Remove "Generate Invoice" button completely**

9.3 Order Detail Expansion

File: [orders/[id]/page.tsx](<file:///c:/Users/kidus/Documents/Projects/printonline-et/app/(cms)/cms/orders/[id]/page.tsx>)

Expand:

1. **Order Items** — per-item total, per-option details, product images, tax inclusive
2. **Customer Profile** — ALL user fields + CTA to customer detail page
3. **Logistics** — full delivery address, sub-city, special instructions, production priority
4. **Design Files** — display uploaded files with preview/download
5. **Status Control** — strict pipeline (Section 10)
6. Remove gimmicks → premium & professional

Files:

File	Action
`components/cms/orders/OrderList.tsx`	**MODIFY** (polish)
`app/(cms)/cms/orders/[id]/page.tsx`	**MAJOR REFACTOR**
`components/cms/orders/OrderCustomerInfo.tsx`	**MAJOR REFACTOR**
`components/cms/orders/OrderFulfillmentInfo.tsx`	**MAJOR REFACTOR**
`components/cms/orders/OrderItemList.tsx`	**MODIFY**

---

10. Order Status Pipeline

10.1 Status Enum (Strict 8-Step)

Already in [types/database.ts](file:///c:/Users/kidus/Documents/Projects/printonline-et/types/database.ts#L718-L726):

```
Order Confirmed → Design Under Review → On Hold → Approved for Production
  → Printing in Progress → Ready for Delivery → Out for Delivery → Delivered
```

10.2 CMS Status Control Rules

Current Status	Allowed Next	Email?
`order_confirmed`	`design_under_review`	☑
`design_under_review`	`on_hold`, `approved_for_production`	☑
`on_hold`	`design_under_review`	☑ (with hold reason)
`approved_for_production`	`printing_in_progress`	☑
`printing_in_progress`	`ready_for_delivery`	☑
`ready_for_delivery`	`out_for_delivery`	☑
`out_for_delivery`	`delivered`	☑
`delivered`	(terminal)	☑ delivery confirmation

- Admin sees dropdown with **only valid next statuses**
- Each change requires **confirmation dialog**
- Each change fires **email notification** to customer
- Each change appends to `status_history` JSONB

Files:

- `app/(cms)/cms/orders/[id]/page.tsx` — **MODIFY** (strict control)
- `lib/validations/cms.ts` — **MODIFY** (status transition rules)
- [OrderStatusTracker.tsx](file:///c:/Users/kidus/Documents/Projects/printonline-et/components/order/OrderStatusTracker.tsx) — **VERIFY**
- [orders/page.tsx (account)](<file:///c:/Users/kidus/Documents/Projects/printonline-et/app/(account)/orders/page.tsx>) — **VERIFY**

---

11. Customer Management Module

11.1 Customer List

Extend [customers/page.tsx](<file:///c:/Users/kidus/Documents/Projects/printonline-et/app/(cms)/cms/customers/page.tsx>):

- Columns: name, email, phone, company, TIN, total orders, last order date
- Search/filter
- Links to detail page

11.2 Customer Detail

Refactor [customers/[id]/page.tsx](<file:///c:/Users/kidus/Documents/Projects/printonline-et/app/(cms)/cms/customers/[id]/page.tsx>):

- Full profile display (all `customer_profiles` fields)
- **Order history** inline — all orders with details
- Edit form (name, phone, TIN, company, address)
- Deactivate/reactivate toggle

Files:

- `app/(cms)/cms/customers/page.tsx` — **REFACTOR**
- `app/(cms)/cms/customers/[id]/page.tsx` — **MAJOR REFACTOR**
- `hooks/data/useCustomers.ts` — **MODIFY** (CRUD mutations)

---

12. Email Integration Master Plan

12.1 Event Matrix

Event	Recipients	Priority
Order Placed	Customer + admin	P0
Welcome (Account Created)	Customer	P0
Password Reset	Customer	P0
Order Status Change	Customer	P0
Order Cancellation	Customer	P0
Delivery Status	Customer	P1
Direct Admin Contact	User	P1
New Order Admin Alert	Admin	P0
System Notifications	Admin	P2

12.2 Infrastructure (already exists)

- [lib/email.ts](file:///c:/Users/kidus/Documents/Projects/printonline-et/lib/email.ts) — Nodemailer transport
- [lib/email-template.ts](file:///c:/Users/kidus/Documents/Projects/printonline-et/lib/email-template.ts) — HTML builder
- `.env.local` — SMTP creds
- [app/api/send-order-email/route.ts](file:///c:/Users/kidus/Documents/Projects/printonline-et/app/api/send-order-email/route.ts) — existing endpoint
- [/email-integration workflow](file:///c:/Users/kidus/Documents/Projects/printonline-et/.agent/workflows/email-integration.md) — full integration guide

12.3 Phased Build

Phase 1 (v3.0): Welcome, Password Reset, Order Confirmation, Cancellation
Phase 2 (v3.1): Status Change, New Order Admin Alert, Admin Contact
Phase 3 (v3.2): Delivery Status, System Notifications

Files to Create:

- `lib/email-templates/welcome.ts`
- `lib/email-templates/password-reset.ts`
- `lib/email-templates/order-confirmation.ts`
- `lib/email-templates/order-cancellation.ts`
- `lib/email-templates/status-update.ts`
- `lib/email-templates/admin-notification.ts`
- `app/api/email/route.ts` — unified email dispatch

---

13. Deferred Modules

13.1 Products → Under Maintenance

- Disable in sidebar with maintenance badge
- `/cms/products` page shows: "Product management is under maintenance. Available in v3.3 alongside Chapa payment integration."

13.2 Categories → Under Maintenance

Same treatment as products.

13.3 Settings → Remove Completely

Remove from sidebar navigation ([CMSSidebar.tsx lines 168-173](file:///c:/Users/kidus/Documents/Projects/printonline-et/components/cms/layout/CMSSidebar.tsx#L168-L173)).

---

14. Admin User Provisioning

Add: `belaynehhenok@gmail.com` as admin

Check admin detection method (likely env var `ADMIN_EMAILS` or DB field) and add accordingly.

---

15. Implementation Priority & Phasing

Phase 1: Critical Data Patches (~5h)

#	Task	Est
1	Pricing matrix migration + seed SQL	1h
2	Frontend pricing logic refactor	1.5h
3	Remove `+` sign from option prices	15min

#	Task	Est
4	Fix flyer options + add sticker lamination	30min
5	Quantity enforcement	45min
6	Rush production cart total fix	30min
7	Remove cart quantity selector	20min

Phase 2: Storefront UX (~4h)

#	Task	Est
8	T&C on checkout	45min
9	Out of stock badges	30min
10	Multiple file upload	1h
11	"I don't have a Design" label	5min
12	Search bar premium functionalization	1.5h

Phase 3: CMS Overhaul (~8h)

#	Task	Est
13	Sidebar refactor (logo/settings/maintenance)	30min
14	Font weight overhaul (all CMS)	1h
15	Animation overhaul (all CMS)	30min
16	Dashboard redesign	1h
17	Products/Categories maintenance page	20min
18	Order list polish + export	30min
19	Cancel order + confirmation dialog	45min
20	Order detail page expansion	2h
21	Order status pipeline control	1h
22	Customer management CRUD + detail	2h

Phase 4: Email Integration (~3.5h)

#	Task	Est
23	Auth emails (welcome + reset)	1h
24	Order confirmation email	30min
25	Order cancellation email	30min

#	Task	Est
26	Status change emails	1h
27	Admin notification emails	30min

Phase 5: Misc (~15min)

#	Task	Est
28	Add admin user	10min
29	Remove "Generate Invoice" button	5min

Total: ~20.5 hours

---

16. File Reference Map

New Files

File	Purpose
`supabase/migrations/010_pricing_matrix.sql`	Pricing matrix table
`supabase/seed/007_pricing_matrix.sql`	All pricing data + option fixes
`components/checkout/TermsCheckbox.tsx`	T&C checkbox
`lib/email-templates/welcome.ts`	Welcome email
`lib/email-templates/password-reset.ts`	Password reset
`lib/email-templates/order-confirmation.ts`	Order confirmation
`lib/email-templates/order-cancellation.ts`	Cancellation
`lib/email-templates/status-update.ts`	Status change
`lib/email-templates/admin-notification.ts`	Admin alerts
`app/api/email/route.ts`	Unified email API

Major Refactors

File	What Changes
`components/product/ProductOrderForm.tsx`	Pricing logic, file upload, quantity, label
`components/cms/layout/CMSSidebar.tsx`	Logo, settings removed, maintenance
`components/cms/layout/CMSHeader.tsx`	Text removed, logo only
`app/(cms)/cms/page.tsx`	Dashboard redesign
`app/(cms)/cms/orders/[id]/page.tsx`	Expanded order detail + status control

File	What Changes
`app/(cms)/cms/customers/[id]/page.tsx`	Full customer detail + history
`context/CartContext.tsx`	Multiple files, priority in total
`app/order-summary/page.tsx`	T&C gate, file upload
`components/shared/SearchBar.tsx`	Premium live search

Minor Modifications

File	What Changes
`components/shared/ProductCard.tsx`	Out of stock badge
`components/product/ProductInfo.tsx`	Out of stock banner
`components/cart/CartItem.tsx`	Remove quantity +/-
`components/layout/CartDrawer.tsx`	Remove quantity editing
`lib/auth.ts`	Email hooks
`hooks/data/useProduct.ts`	Include pricing_matrix
`hooks/data/useCustomers.ts`	CRUD mutations
`types/index.ts`	pricing_matrix in types
`lib/validations/cms.ts`	Status transition rules
`components/cms/orders/OrderList.tsx`	Polish, columns, export
`components/cms/orders/OrderCustomerInfo.tsx`	Full customer display
`components/cms/orders/OrderFulfillmentInfo.tsx`	Logistics expansion
`components/cms/orders/OrderItemList.tsx`	Price breakdowns

> **This document serves as the single source of truth for PrintOnline.et v3.0 implementation.**
> **All tasks should reference this document for context, specifications, and file locations.**

Last updated: 2026-03-11T16:08 EAT